

# Methods of Integration

## **Complex Analysis**

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### Example

*Use the principal-value method to evaluate the integral*

$$\int_{\mathbb{R}^+} dx \frac{\sin x}{x}.$$

### Example

Let  $\Gamma$  be a positively oriented, simple, closed path containing  $i$  in its interior and  $-i$  in its exterior. Evaluate

$$\int_{\Gamma} dz \frac{1}{z^2 + 1}.$$

Compute the same integral when  $\Gamma$  contains both  $i$  and  $-i$  in its interior.

### Example

Let  $\Sigma$  be a simple closed positively oriented path such that  $1, -i$ , and  $i$  are in the interior of  $\Sigma$  and  $-1$  is in the exterior of  $\Sigma$ . Find

$$\oint_{\Sigma} dz \frac{1}{z^4 - 1}.$$

### Example

Let  $\Sigma$  be a simple closed positively oriented path such that 1 is in the interior of  $\Sigma$  and  $-1$  is in the exterior of  $\Sigma$ . Evaluate

$$\oint_{\Sigma} dz \frac{\sin \pi z}{z^2 - 1}.$$

### Example

*Compute the integral*

$$\int_{\mathbb{R}} dx \frac{x^2}{(x^2 + 1)(x^2 + 9)}.$$

### Example

Proceed as indicated in the steps that follow.

- a) Let  $k$  be an integer. Show that  $\text{Res}(\cot \pi z, k) = \frac{1}{\pi}$ .
- b) Let  $C$  be the positively oriented rectangular path shown below. Evaluate

$$\oint_C dz \frac{\cot \pi z}{z^4 + 1}.$$

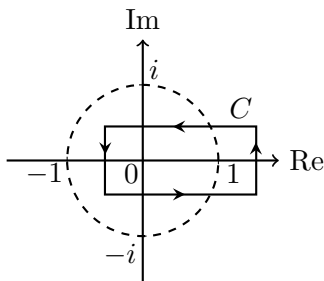


Figure 1: Rectangular path  $C$

## Remark

We consider integrals of the form

$$\int_0^{2\pi} d\theta F(\cos \theta, \sin \theta),$$

where  $F$  is a rational function whose denominator does not vanish (as the composition with  $\cos$  and  $\sin$ ) on the interval  $[0, 2\pi]$ . Using  $z = e^{i\theta}$  and the substitutions

$$\cos \theta = \frac{1}{2} \left( z + \frac{1}{z} \right), \quad \sin \theta = \frac{1}{2i} \left( z - \frac{1}{z} \right), \quad -i \frac{dz}{z} = d\theta,$$

we will transform the (real) integral into a contour integral and apply the developed methods of contour integration. If  $n$  is an integer, we have the following pair of useful identities (due to De Moivre's formula)

$$\cos n\theta = \frac{1}{2} \left( z^n + \frac{1}{z^n} \right) \quad \text{and} \quad \sin n\theta = \frac{1}{2i} \left( z^n - \frac{1}{z^n} \right).$$

### Example

*Evaluate*

$$\int_0^{2\pi} d\theta \frac{1}{10 + 8 \cos \theta}.$$

Example

*Evaluate*

$$\int_0^{2\pi} d\theta \frac{\cos 2\theta}{2 + \cos \theta}$$

## Example

*Provide the details in the calculation shown in the picture below.*

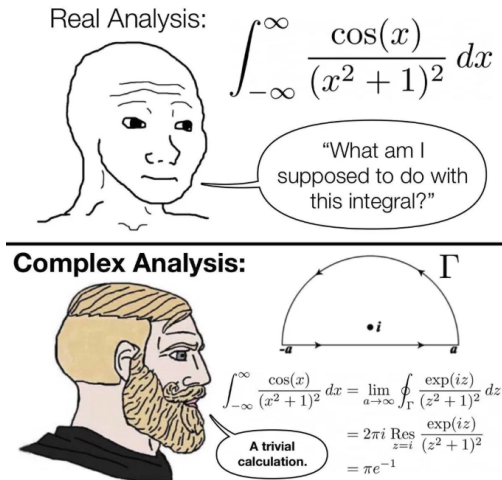


Figure 2: Complex Analysis integral meme

### Example

Evaluate the **Gaussian integral**

$$\int_{\mathbb{R}} dx e^{-x^2}.$$

Compute a variation thereof, namely

$$\int_{\mathbb{R}} dx e^{iax-bx^2}, \quad a, b \in \mathbb{R}, b > 0.$$

### Example

*Use the previous result to deduce a formula for*

$$\int_{\mathbb{R}} dx \exp(-ax^2 + bx + c).$$

Example

*Evaluate*

$$\int_{\mathbb{R}^+} dx \frac{1}{x^3 + 1}.$$

Example

*Evaluate*

$$I = \int_{\mathbb{R}} dx \frac{x^2}{x^4 + 1}.$$

### Example

Let  $\alpha > 1$ . Establish the identity

$$\int_{\mathbb{R}^+} dx \frac{1}{x^\alpha + 1} = \int_{\mathbb{R}} dx \frac{e^x}{e^{\alpha x} + 1} = \frac{\pi}{\alpha \sin \frac{\pi}{\alpha}}.$$